TIANYU WANG

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Education

Georgia Institute of Technology (GT), Atlanta, GA Jan 2021 - Dec 2025
PhD in Robotics
Advisor: Daniel I. Goldman
Thesis: Mechanically intelligent elongate limbless robots for locomotion in complex land and water environments
Committee: Daniel I. Goldman, Hang Lu, David Hu, Tony Chen, Howie Choset

Carnegie Mellon University (CMU), Pittsburgh, PA Aug 2018 - May 2020
MS in Mechanical Engineering
Advisor: Howie Choset
Thesis: Directional compliance in obstacle-aided navigation for snake robots

Shanghai Jiao Tong University (SJTU), Shanghai, China Sep 2014 - Aug 2018
The University of Michigan-Shanghai Jiao Tong University Joint Institute
BS in Electrical and Computer Engineering
Research advisor: Guoying Gu

Experience

Postdoctoral Research Fellow, Complex Rheology and Biomechanics Lab, GT Jan 2026 - Present
Advisor: Daniel I. Goldman

Graduate Research Assistant, Complex Rheology and Biomechanics Lab, GT Jan 2021 - Dec 2025
Advisor: Daniel I. Goldman

Research Staff, Biorobotics Lab, CMU May 2020 - Dec 2020
Advisor: Howie Choset

Graduate Research Assistant, Biorobotics Lab, CMU Sep 2019 - May 2020
Advisor: Howie Choset

Honors and Awards

Rising Star in Mechanical Engineering 2026
One of 30 early-career researchers selected nationally in mechanical engineering.

RSS Pioneer 2026
One of 30 early-career robotics researchers selected worldwide.

Trailblazers in Engineering Fellow 2026
One of 22 early-career scholars selected nationally in engineering.

Best Teaser Talk, Workshop on Miniature Multiterrain and Multimodal Locomotion, IROS 2026

MESH Workshop Travel Grant 2026

NSF CPS (Cyber-Physical Systems) Rising Star 2025
One of 30 PhD students and postdocs selected in cyber-physical systems.

Georgia Tech School of Mechanical Engineering Ph.D. Research Excellence Award (Glenn Fellowship) 2025

APS FGSA Graduate Research Excellence Travel Award 2025

Topping Fellowship (Paul Yopp Fellowship) 2025
Recognizing academic achievement and undergrad student mentoring

Georgia Tech Center for Promoting Inclusion and Equity in Science Research Experience Award 2024

Topping Fellowship (Paul Yopp Fellowship) 2023, 2024
Recognizing academic achievement

Georgia Tech IRIM Robotics Research Showcase Best Poster Award 2022

Robotics: Science and Systems (RSS) Best Paper Finalist ([C5]) 2021

SJTU Academic Excellence Scholarship 2015, 2016, 2017, 2018

Silver Medal in Advanced Vision Challenge, RoboCup China Open 2016

Covidien Scholarship 2014

Publications

Google Scholar page Total citations (October 2026): 840 H-index: 13

Selected publications include Science, Science Robotics, PNAS, IJRR.

* Authors contributed equally

Preprints

[R3] **T. Wang***, M. Fernandez*, G. Tunnicliffe, N. Cornell, J. Duong, D. Dortilus, Z. Xu, P. Meza, S. Lublinsky, D. Parikh, J. Lin, E. Grace, D. I. Goldman. Three-dimensional hydro-cluttered locomotion by an undulatory robot. ***Science Robotics*** *(In Revision)*. 2026.

[R2] B. Chong*, **T. Wang***, K. Diaz, C. Pierce, E. Erickson, J. Whitman, Y. Deng, E. Flores, R. Fu, J. He, J. Lin, H. Lu, G. Sartoretti, H. Choset, D. I. Goldman. The omega turn: A general turning template for elongate robots. 2025.

[R1] B. Chong, J. He, D. Irvine, **T. Wang**, E. Flores, D. Soto, J. Lin, Z. Xu, V. Nienhusser, G. Blekherman, D. I. Goldman. Robust control for multi-legged elongate robots in noisy environments. 2025.

Refereed Journal Papers

[J17] X. Yang, X. Ding, **T. Wang**, D. I. Goldman, S. Li, B. Deng. Toward Mechanical Intelligence in Robotics. *Advanced Robotics Research*, 2026.

[J16] J. Lin*, **T. Wang***, B. Chong, M. Fernandez, Z. Xu, D. I. Goldman. Optimal swimming with body compliance in an overdamped medium. *Physical Review Applied*, 2026.

[J15] C. J. Pierce*, **T. Wang***, D. Kalinin, A. Zangwill, D. I. Goldman. Aubry-André Localization Transition for an Active Undulator. *The Proceedings of the National Academy of Sciences (PNAS)*, 2026.

[J14] G. Volpe, N. A. Araújo, ..., **T. Wang**, *et al.* Roadmap for Animate Matter. *Journal of Physics: Condensed Matter*, 2025.

[J13] B. Liu*, **T. Wang***, D. Kerimoglu, V. Kojouharov, F. L. Hammond III, D. I. Goldman. Robust self-propulsion in sand using simply controlled vibrating cubes. *Frontiers in Robotics and AI*, 2024.

[J12] **T. Wang***, C. J. Pierce*, V. Kojouharov, B. Chong, K. Diaz, H. Lu, D. I. Goldman. Mechanical intelligence simplifies control in terrestrial limbless locomotion. ***Science Robotics***, 2023.

[J11] B. Chong, J. He, D. Soto, **T. Wang**, D. Irvine, G. Blekherman, D. I. Goldman. Multi-legged matter transport: a framework for locomotion on noisy landscapes. ***Science***, 2023.

[J10] B. Chong, **T. Wang**, B. Lin, S. Li, P. Muthukrishnan, J. He, D. Irvine, H. Choset, G. Blekherman and D. I. Goldman. Optimizing contact patterns for robot locomotion via geometric mechanics. *The International Journal of Robotics Research (IJRR)*, 2023.

[J9] B. Chong, J. He, S. Li, E. Erickson, K. Diaz, **T. Wang**, D. Soto, D. I. Goldman. Self-propulsion via slipping: Frictional swimming in multilegged locomotors. *The Proceedings of the National Academy of Sciences (PNAS)*, 2023.

[J8] S. Li*, **T. Wang***, V. Kojouharov, J. McInerney, E. Aydin, Y. Ozkan-Aydin, D. I. Goldman, D. Z. Rocklin. Robotic swimming in curved space via geometric phase. *The Proceedings of the National Academy of Sciences (PNAS)*, 2022.

[J7] B. Chong, **T. Wang**, E. Erickson, P. J. Bergmann, D. I. Goldman. Coordinating tiny limbs and long bodies: Geometric mechanics of lizard terrestrial swimming. *The Proceedings of the National Academy of Sciences (PNAS)*, 2022.

[J6] B. Chong, Y. O. Aydin, J. M. Rieser, G. Sartoretti, **T. Wang**, J. Whitman, A. Kaba, E. Aydin, C. McFarland, H. Choset and D. I. Goldman. A general locomotion control framework for serially connected multi-legged robots. *Bioinspiration & Biomimetics*, 2022.

[J5] B. Chong*, **T. Wang***, J. Rieser, B. Lin, A. Kaba, G. Blekherman, H. Choset and D. I. Goldman. Frequency modulation of body waves to improve performance of sidewinding robots. *The International Journal of Robotics Research (IJRR)*, 2021.

[J4] **T. Wang***, B. Lin*, B. Chong, J. Whitman, M. Travers, D. I. Goldman, G. Blekherman, H. Choset. Reconstruction of backbone curves for snake robots. *IEEE Robotics and Automation Letters (RA-L)*, 2021.

[J3] **T. Wang***, L. Ge*, and G. Gu. Programmable design of soft pneu-net actuators with oblique chambers can generate coupled bending and twisting motions. *Sensors and Actuators A: Physical*, 2018.

[J2] L. Ge*, **T. Wang***, N. Zhang, and G. Gu. Fabrication of soft pneumatic network actuators with oblique chambers. *Journal of Visualized Experiments*, 2018.

[J1] S. Wei, **T. Wang**, and G. Gu. Design of a soft pneumatic robotic gripper based on fiber-reinforced actuator. *Chinese Journal of Mechanical Engineering*, 2017.

Refereed Conference Papers

[C14] D. Dortilus*, **T. Wang***, G. Tunnicliffe, M. Fernandez, D. I. Goldman. Morphing MILR: Design and control of a cable-driven limbless robot with rolling joints for maneuvering in complex environments. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2026.

[C13] Z. Xu, J. He, D. Aydin, M. Taylor, **T. Wang**, J. Lin, W. Dyar, D. I. Goldman. A robust antenna provides tactile feedback in a multi-legged robot. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2026.

[C12] M. Fernandez*, **T. Wang***, G. Tunnicliffe, D. Dortilus, P. Gunnarson, J. O. Dabiri, D. I. Goldman. AquaMILR+: Design of an untethered limbless robot for complex aquatic terrain navigation. *IEEE International Conference on Robotics and Automation (ICRA)*, 2025.

[C11] **T. Wang***, N. Mankame*, M. Fernandez, V. Kojouharov, D. I. Goldman. AquaMILR: Mechanical intelligence simplifies control of undulatory robots in cluttered fluid environments. *IEEE International Conference on Robotics and Automation (ICRA)*, 2025.

[C10] M. Iaschi, B. Chong, **T. Wang**, J. Lin, J. He, D. Soto, Z. Xu, D. I. Goldman. Addition of a peristaltic wave improves multi-legged locomotion performance on complex terrains. *IEEE International Conference on Robotics and Automation (ICRA)*, 2025.

[C9] E. Teder, B. Chong, J. He, **T. Wang**, M. Iaschi, D. Soto, D. I. Goldman. Effective self-righting strategies for elongate multi-legged robots. *IEEE International Conference on Robotics and Automation (ICRA)*, 2025.

[C8] V. Kojouharov*, **T. Wang***, M. Fernandez, J. Maeng, D. I. Goldman. Anisotropic body compliance facilitates robotic sidewinding in complex environments. *IEEE International Conference on Robotics and Automation (ICRA)*, 2024.

[C7] B. Chong, **T. Wang**, D. Irvine, V. Kojouharov, B. Lin, H. Choset, D. I. Goldman, G. Blekherman. Gait design for limbless obstacle aided locomotion using geometric mechanics. *Robotics: Science and Systems (RSS)*, 2023.

[C6] **T. Wang***, B. Chong*, Y. Deng, R. Fu, H. Choset, D. I. Goldman. Generalized omega turn gait enables agile limbless robot turning in complex environments. *IEEE International Conference on Robotics and Automation (ICRA)*, 2022.

[C5] B. Chong, **T. Wang**, B. Lin, S. Li, G. Blekherman, H. Choset, D. I. Goldman. Moving sidewinding forward: optimizing contact patterns for limbless robots via geometric mechanics. *Robotics: Science and Systems (RSS)*, 2021. **Best Paper Award Finalist**

[C4] G. Sartoretti, **T. Wang**, G. Chuang, Q. Li, H. Choset. Autonomous decentralized shape-based navigation for snake robots in dense environments. *IEEE International Conference on Robotics and Automation (ICRA)*, 2021.

[C3] **T. Wang***, B. Chong*, K. Diaz, J. Whitman, H. Lu, M. Travers, D. I. Goldman and H. Choset. The omega turn: a biologically-inspired turning strategy for elongated limbless robots. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2020.

[C2] B. Chong, **T. Wang**, J. Rieser, A. Kaba, H. Choset and D. I. Goldman. Frequency modulation of body waves to improve performance of limbless robots. *Robotics: Science and Systems (RSS)*, 2020.

[C1] **T. Wang**, J. Whitman, M. Travers, and H. Choset. Directional compliance in obstacle-aided navigation for snake robots. *American Control Conference (ACC)*, 2020.

Thesis

[T2] **T. Wang**. Mechanically intelligent elongate limbless robots for locomotion in complex land and water environments. Ph.D. Thesis, Georgia Institute of Technology.

[T1] **T. Wang**. Directional compliance in obstacle-aided navigation for snake robots. M.S. Thesis, Carnegie Mellon University.

Conference Abstracts/Presentations/Posters

First authored only. Full list refer to my Google scholar page

[A17] **T. Wang**, M. Fernandez, G. Tunnicliffe, D. Dortilus, C. Pierce, D. I. Goldman. Mechanical and computational intelligence for agile and robust limbless robotic locomotion in complex aquatic environments. *APS Global Physics Summit*, 2025.

[A16] **T. Wang**, M. Fernandez, G. Tunnicliffe, D. Dortilus, D. I. Goldman. Mechanically intelligent undulatory robotic locomotion in complex aquatic environments. *SICB Annual Meeting*, 2025.

[A15] **T. Wang**, B. Chong, A. Bhumkar, V. Kojouharov, C. Pierce, D. I. Goldman. Gait design and mechanical intelligence facilitate open-loop limbless obstacle aided locomotion. *American Physical Society March Meeting*, 2024.

[A14] **T. Wang**. Mechanical intelligence in undulatory locomotors. *Gordon Research Conference: Robotics*, 2024.

[A13] **T. Wang**. Mechanical intelligence in undulatory locomotors. *Gordon Research Seminar: Robotics*, 2024.

[A12] **T. Wang**, N. Mankame, A. Bhumkar, V. Kojouharov, C. Pierce, D. I. Goldman. Physical intelligence aids limbless locomotion in cluttered aquatic environments. *SICB Annual Meeting*, 2024.

[A11] **T. Wang**, C. Pierce, V. Kojouharov, K. Diaz, B. Chong, H. Lu, D. I. Goldman. Lattice transport via mechanical intelligence in undulatory locomotors. *American Physical Society March Meeting*, 2023.

[A10] **T. Wang**, V. Kojouharov, C. Pierce, K. Diaz, B. Chong, V. Zborovsky, D. I. Goldman. Robophysical modeling reveals the role of passive body mechanics in *C. elegans* locomotion. *SICB Annual Meeting*, 2023.

[A9] **T. Wang**, V. Kojouharov and D. I. Goldman. A novel limbless robot for complex terrain navigation via passive mechanical interactions. *GT IRIM Robotics Research Showcase*, 2022. **Best Poster Award**

[A8] **T. Wang**, D. Z. Rocklin, D. L. Hu and D. I. Goldman. Experiment and analysis of limbless robot locomotion in heterogeneous environment from a macroscopic perspective. *GT IRIM Robotics Research Showcase*, 2022.

[A7] **T. Wang**, B. Chong, Y. Deng, R. Fu, H. Choset, D. I. Goldman. Worm omega turn modeling and its limbless robot implementation via geometric mechanics. *American Physical Society March Meeting*, 2022.

[A6] **T. Wang**, M. C. Maisonneuve, K. Diaz, P. E. Schiebel, D. I. Goldman. Complex terrain navigation via passive mechanical interactions in a novel limbless robot. *SICB Annual Meeting*, 2022.

[A5] **T. Wang**, B. Chong, J. He, K. Diaz, E. Erickson, D. I. Goldman. Robophysical modeling of the coordination between body undulation and leg movement in centipedes. *SICB Annual Meeting*, 2022.

[A4] **T. Wang**, B. Lin, B. Chong, J. Whitman, M. Travers, D. I. Goldman, H. Choset, G. Blekherman. Reconstruction of Backbone Curves for 3-D Locomotion of Limbless Robots. *American Physical Society March Meeting*, 2021.

[A3] **T. Wang**, B. Chong, K. Diaz, J. Whitman, H. Lu, M. Travers, D. I. Goldman and H. Choset. A Biologically Inspired Omega-Shaped Turning Gait for Elongated Limbless Robots. *American Physical Society March Meeting*, 2021.

[A2] **T. Wang**, B. Chong, K. Diaz, J. Whitman, H. Lu, M. Travers, D. I. Goldman and H. Choset. The omega turn:

a biologically-inspired turning strategy for elongated limbless robots. *Workshop: Robotics-Inspired Biology in 2020 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2020.

[A1] **T. Wang**, J. Whitman, M. Travers, and H. Choset. Directional compliance in snake robot obstacle-aided locomotion. *American Physical Society March Meeting*, 2020.

Patents

[P4] **T. Wang**, D. Dortilus, and D. I. Goldman. Swiveling cable-driven joint. *US Patent (filed 2025)*.

[P3] **T. Wang**, M. Fernandez, and D. I. Goldman. Compact buoyancy control assembly for underwater robotic systems or equipment. *US Patent (filed 2024)*.

[P2] **T. Wang**, V. Kojouharov, and D. I. Goldman. Devices and systems for locomoting diverse terrain and methods of use. *US20250136206A1*, 2024.

[P1] G. Gu, L. Dong, **T. Wang**, and X. Zhu. A kind of device and its application method of the control of 3D printing device location. *CN108081596A*, 2017.

Invited Talks

Department of Mechanical Engineering, Pennsylvania State University (Host: Baxi Chong)	Oct 2026
RSS Pioneer Workshop	Jul 2026
The Mechano-Computation for Expanding Scientific Horizons Network Kick-off Workshop	May 2026
CMU Safe AI Lab	Apr 2024
Seattle Robotics Society	Jun 2023
NSF Physics of Living Systems (PoLS) Seminar Series	Oct 2022
Georgia Tech RoboGrads Student Seminar Series	Apr 2022
NSF Physics of Living Systems (PoLS) Seminar Series	Feb 2022

Teaching Experience

ME 456 Introduction to Robotics, Pennsylvania State University Fall 2026
Guest Instructor
Delivered a guest lecture on robot locomotion.

Robophysics Summer Bootcamp, Georgia Tech Summer 2025
Instructor
Delivered lectures and supervised laboratory sessions.

Robophysics Summer Bootcamp, Georgia Tech Summer 2024
Instructor
Delivered lectures.

VM467 Introduction to Robotics, Shanghai Jiao Tong University Spring 2018
Teaching Assistant
Instructor: Prof. Yu Zheng

VE216 Signals and Systems, Shanghai Jiao Tong University Spring 2017
Teaching Assistant
Instructor: Prof. Mohamed Atef

Mentored Students

Mentored 15+ undergraduate and graduate researchers; multiple award recipients and PhD placements.

Velin (Venny) Kojouharov (ME Undergrad at GT, 2021–2024, proceeded to ME PhD at Stanford)
Co-authored publications: [J12][J8][C8][C7][P2]; 10 conference abstracts/posters/presentations

Awards: *Georgia Tech President's Undergraduate Research Award, Millennium Fellowship, Barry Goldwater Scholarship, Astronaut Scholarship, ThinkSwiss Research Scholarship, Churchill Scholarship, Love Family Foundation Award, Knight-Hennessy Scholarship, NSF Graduate Research Fellowship*

Matthew Fernandez (ME Undergrad at GT, 2023–2026, proceeded to ME PhD at MIT)

Co-authored publications: [J16][C14][C12][C11][C8][P3]; 6 conference abstracts/posters/presentations

Awards: *Astronaut Scholarship, Georgia Tech President's Undergraduate Research Award, Barry Goldwater Scholarship, ThinkSwiss Research Scholarship, NSF Graduate Research Fellowship*

Jianfeng Lin (QBioS PhD at GT, 2023–present)
Co-authored publications: [J16][C13][C10]; 4 conference abstracts/posters/presentations

Zhaochen (James) Xu (Summer intern at GT, 2023, QBioS PhD at GT, 2024–present)
Co-authored publications: [J16][C13][C10]; 2 conference abstracts/posters/presentations

Dmitri Kalinin (Physics Undergrad at GT, 2023–present)
Co-authored publications: [J15]; 2 conference abstracts/posters/presentations

Galen Tunnicliffe (ME Undergrad & Robotics Master at GT, 2024–2026)
Co-authored publications: [C14][C12]; 4 conference abstracts/posters/presentations

Awards: *Georgia Tech President's Undergraduate Research Award*

Donoven Dortilus (ME Undergrad & Robotics Master at GT, 2024–2026)
Co-authored publications: [C14][C12][P4]; 4 conference abstracts/posters/presentations

Nikolas Cornell (ME Undergrad at GT, 2025–present)
Co-authored publications: [R3]

Justin Duong (ME Undergrad at GT, 2025–present)
Co-authored publications: [R3]

Miles Schuler (ME Undergrad at GT, 2025–present)

Darsh Parikh (ECE Master at GT, 2025–present)
Co-authored publications: [R3]

Nishanth Mankame (ME Master at GT, 2023–2024, proceeded to Johns Hopkins University Applied Physics Laboratory as Robotics Engineer)
Co-authored publications: [C11]; 2 conference abstracts/posters/presentations

Anushka Bhumkar (ME Undergrad at GT, 2023–2024, proceeded to International Motors, LLC as Design Engineer)

Co-authored publications: 2 conference abstracts/posters/presentations

Malhar Kamat (ME Undergrad at GT, 2023)

Valerie Zborovsky (ME Undergrad at GT, 2022)
Co-authored publications: 2 conference abstracts/posters/presentations

Zach James (High school student, 2022, proceeded to Undergrad student at Columbia University)

Jin Suh (Summer Scholar at CMU, 2020, proceeded to MS student at UMich, Ann Arbor)

Gabriel Chuang (Undergrad at CMU, 2019–2020, proceeded to Research Staff at Duke University)
Co-authored publications: [C4]

Emmaline Mai (Undergrad at CMU, 2019–2020)

Professional Associations

Professional Societies

Member, Institute of Electrical and Electronics Engineers (IEEE)

Member, American Society of Mechanical Engineers (ASME)

Member, American Physical Society (APS)

Member, Society for Integrative and Comparative Biology (SICB)

Academic Activities and Services

Editorial and Professional Service

Editorial Review Board, International Journal of Advanced Robotic Systems

Workshop Organizer

Towards Agility and Robustness: Mechanical Intelligence in Robotics, Biology, and Smart Materials, IEEE International Conference on Robotics and Automation (ICRA), Atlanta, USA, May 2025. Link

Robophysics Bootcamp, NSF Physics of Living System Student Research Network, Atlanta, GA, July 2025.

Robophysics Bootcamp, NSF Physics of Living System Student Research Network, Atlanta, GA, July 2024.

Agile Movements II: Animal Behavior, Biomechanics, and Robot Devices, IEEE International Conference on Robotics and Automation (ICRA), Yokohama, Japan, May 2024. Link

Agile Movements: Animal Behavior, Biomechanics, and Robot Devices, IEEE International Conference on Robotics and Automation (ICRA), London, the United Kingdom, May 2023. Link

Conference Chair

Session Chair, Sessions *"Legged locomotion over difficult ground"* and *"Walking and rolling"*, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Pittsburgh, PA, United States, September 2026.

Discussion Leader, Session *"Improved Performance Due to a Combination of Hard and Soft Elements"*, Gordon Research Seminar, Ventura, CA, United States, January 2024.

Reviewer (Journals)

Science Robotics

The International Journal of Robotics Research (IJRR)

IEEE Transactions on Robotics (T-RO)

IEEE Robotics & Automation Letters (RA-L)

Soft Robotics

Device (Cell press)

Nonlinear Dynamics

Control Engineering Practice

ACM Transactions on Graphics

Reviewer (Conferences)

IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)

IEEE Conference on Robotics and Automation (ICRA)

IEEE-RAS International Conference on Soft Robotics (RoboSoft)

American Control Conference (ACC)

ACM SIGGRAPH

Award Committee Service

2025 Georgia Tech Undergraduate Research Symposium

2025 Georgia Tech President's Undergraduate Research Award

Press

Georgia Tech Research News 2024
Worms Inspire Wiggly Robots That Navigate All Landscapes Article Video

Georgia Tech Research News 2023
Scurrying Centipedes Inspire Many-Legged Robots That Can Traverse Difficult Landscapes Article Video

Georgia Tech Research News 2022
Robotic Motion in Curved Space Defies Standard Laws of Physics Article Video

Georgia Tech Research News 2022
Tiny Limbs and Long Bodies: Coordinating Lizard Locomotion Article Video

Freethink 2019
These Search and Rescue Robots Could Save Your Life Article Video